

TECHNICAL EVALUATION REPORT

MULTI-MODEL DEEP LEARNING BENCHMARKING FOR BRAIN TUMOR DIAGNOSIS

Project Artifact

Performance Evaluation & Model Selection Dossier

System Module

AI/ML Inference Pipeline Backend

Project Context

Final Year Academic Project (FYP)

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1. EXECUTIVE SUMMARY

This technical evaluation report delivers a rigorous empirical assessment of four prominent convolutional neural network (CNN) architectures trained for autonomous multi-class brain tumor classification. The primary objective is to select the most clinically viable champion model for production deployment within the web application backend. Models evaluated include VGG16, ResNet50, DenseNet121, and EfficientNetB0. Based on exhaustive testing across **1,311** clinical MRI scans, **VGG16** has been designated as the optimal architecture, achieving a global accuracy of **98.78%** and establishing a near-perfect diagnostic profile.

2. BENCHMARKING METHODOLOGY & ENVIRONMENT

To guarantee absolute empirical parity, all models were evaluated under identical execution constraints using a standardized cross-platform evaluation pipeline.

- **Target Classes:** Glioma, Meningioma, Pituitary Tumor, and No Tumor (Healthy Scans).
- **Validation Dataset Size:** n = 1,311 total independent structural MRI frames.
- **Core Evaluation Vectors:** Global Categorical Accuracy, Macro Precision, Macro Recall, Macro-averaged F1-Score, and CPU/GPU Inference Latency (measured as milliseconds elapsed per individual image execution loop).

3. CORE EMPIRICAL FINDINGS

Feature	VGG16	ResNet50	DenseNet121	EfficientNetB0
Total Layers	16	50	121	237
Total Parameters	≈138 Million	≈25.6 Million	≈8 Million	≈5.3 Million
Model Size (Approx.)	528 MB	98 MB	33 MB	29 MB
Core Architecture Concept	Plain Sequential Layers	Skip Connections (Residual)	Dense Connectivity (Feature Reuse)	Compound Scaling (Width, Depth, Resolution)
Computational Efficiency	Low (Slow)	High (Fast)	Moderate-High	Exceptionally High (Super Fast)
Primary Advantage	Deep feature extraction	Solves Vanishing Gradient	Maximizes feature reuse, low params	Highly optimized for lightweight deployment

The consolidated quantitative benchmark data captured during the pipeline run is structured below.

3.1 Aggregated Performance Comparison The high-level statistical evaluation ranks the systems based on their mathematical alignment to the validation ground truths.

Model Architecture	Global Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)	Inference Latency (ms)	Operational Status
VGG16	0.9878	0.9878	0.9878	0.9878	645.70	Selected
ResNet50	0.7559	0.7469	0.7559	0.7427	185.55	Baseline Rejected
DenseNet121	0.7460	0.8096	0.7460	0.7192	191.33	Baseline Rejected
EfficientNetB0	0.4912	0.5290	0.4912	0.4912	549.86	Baseline Rejected

Table 1: Consolidated Deep Learning Model Performance Summary

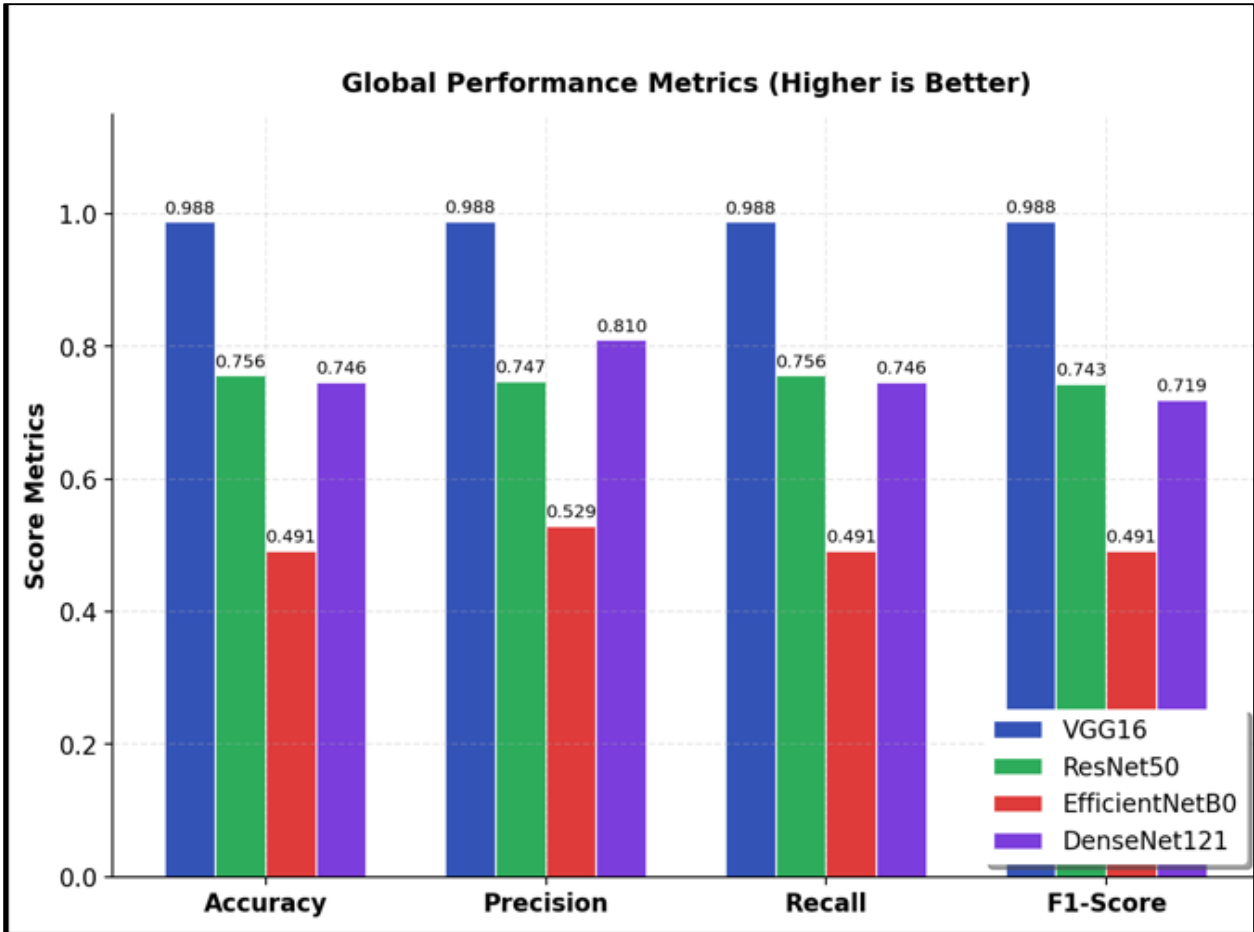


Figure 1: Comparative analysis of global evaluation metrics (Accuracy, Precision, Recall, and F1-Score) across the four deep learning model architectures.

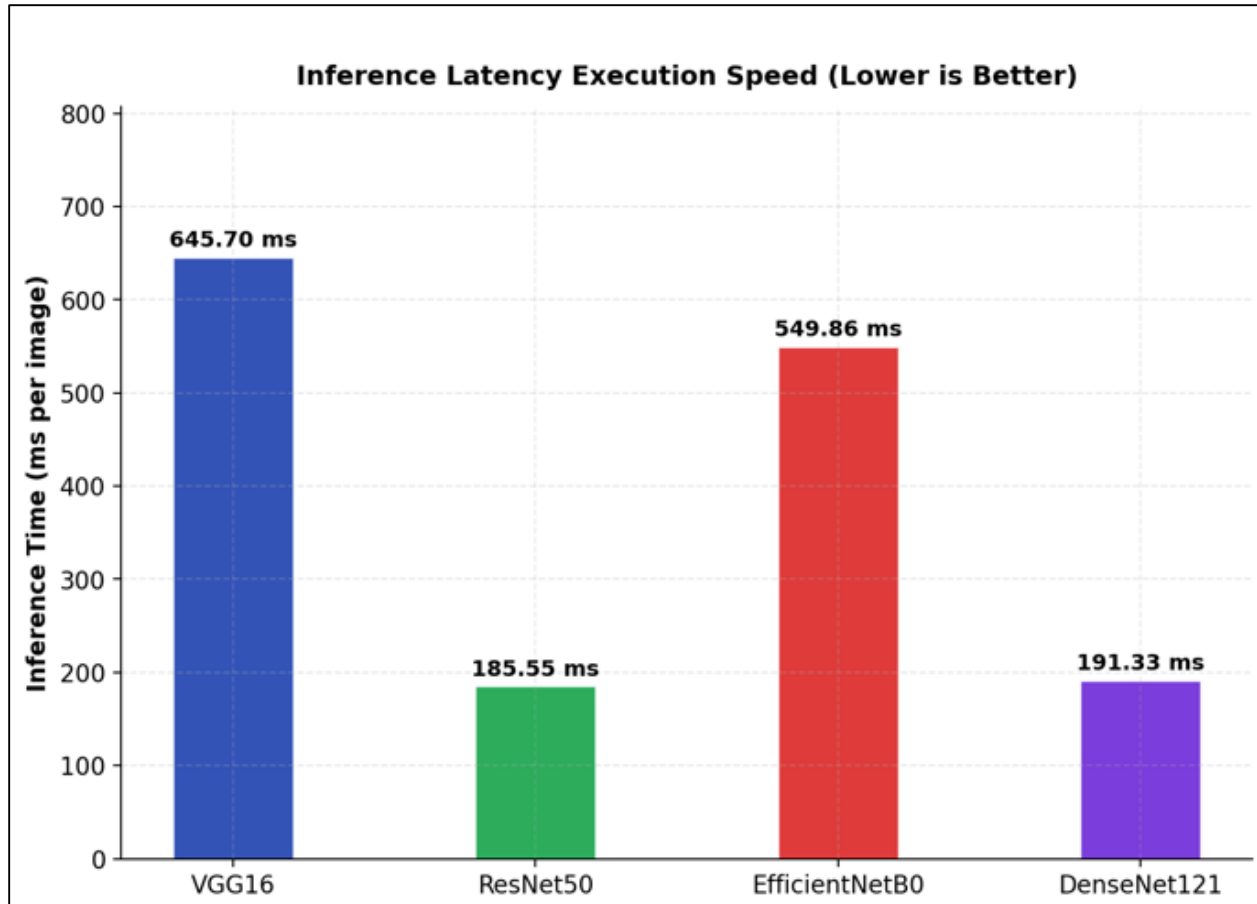


Figure 2: Computational efficiency profile benchmarking the hardware inference latency (measured in milliseconds per image) across the evaluated deep learning frameworks.

3.2 Categorical Diagnostic Precision Breakdowns

To ensure the models do not introduce dangerous biases toward a specific tumor type, class-wise profiles were extracted for every target architecture.

Model Name	Performance Layer	Glioma (n=300)	Meningioma (n=306)	No Tumor (n=405)	Pituitary (n=300)
VGG16	Precision	0.9865	0.9742	0.9951	0.9933
	Recall	0.9767	0.9869	0.9951	0.9900
	F1-Score	0.9816	0.9805	0.9951	0.9917
ResNet50	Precision	0.7637	0.6277	0.8246	0.7468
	Recall	0.6033	0.4739	0.9284	0.9633
	F1-Score	0.6741	0.5400	0.8734	0.8413
DenseNet121	Precision	0.7169	0.9684	0.6574	0.9457
	Recall	0.9200	0.3007	0.9901	0.6967
	F1-Score	0.8058	0.4589	0.7901	0.8023
EfficientNetB0	Precision	0.4489	0.2996	0.7967	0.4816
	Recall	0.6733	0.2614	0.4741	0.5667
	F1-Score	0.5387	0.2792	0.5944	0.5207

Table 2: Micro-Level Class Performance Distribution Matrix

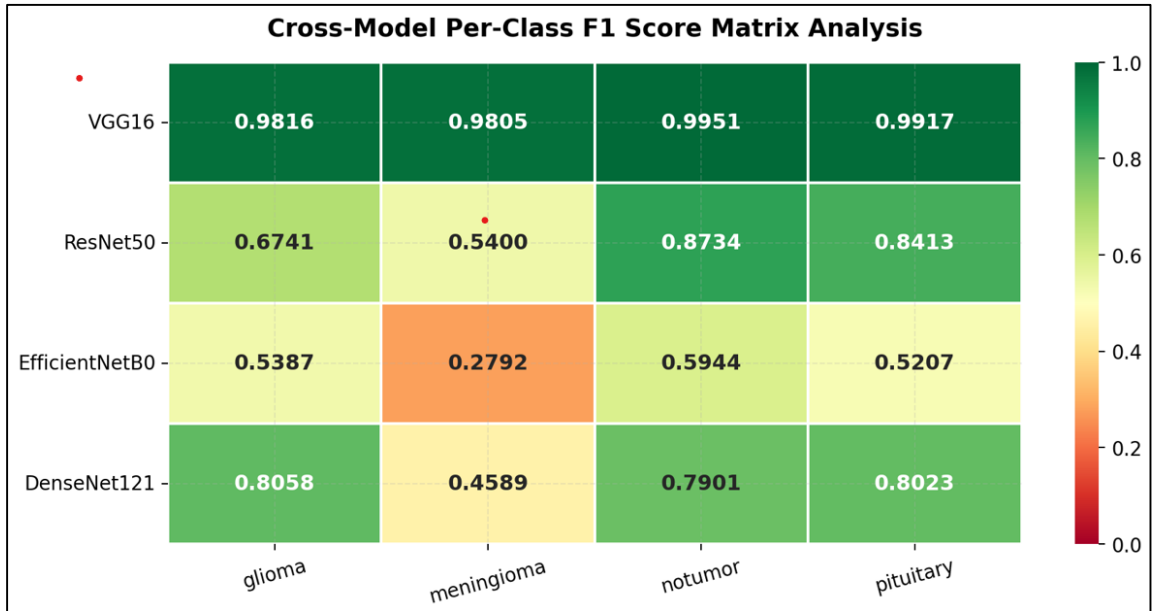


Figure 3: Heatmap matrix distributing the cross-model per-class F1-scores across all targeted tumorous and non-tumorous clinical categories.

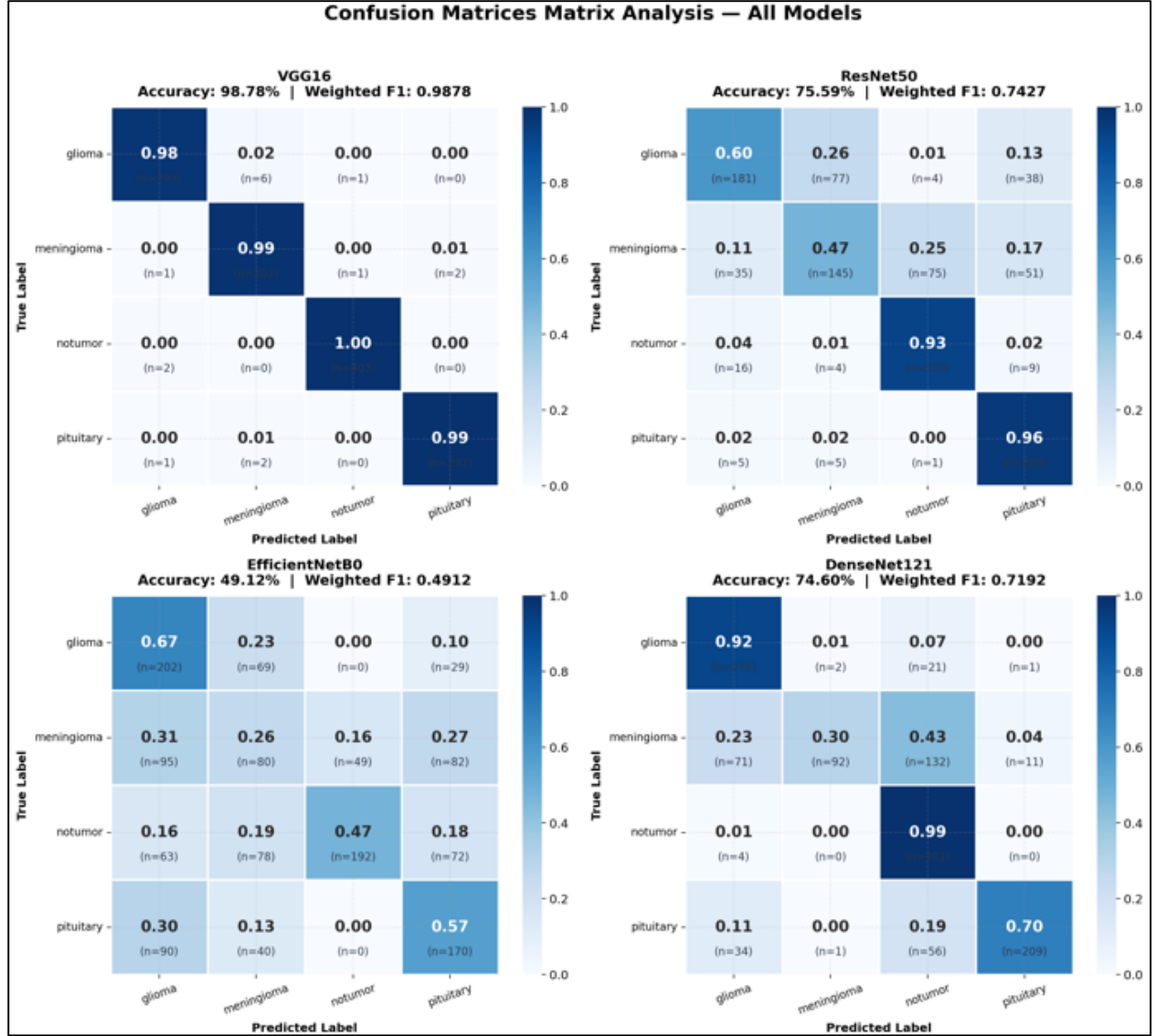


Figure 4: Multi-model normalized confusion matrices evaluating the precise target breakdown of true labels versus predicted labels across the entire validation dataset.

4. ARCHITECTURAL PERFORMANCE ANALYSIS

1. **VGG16 Empirical Domination:** VGG16 vastly outperformed deeper networks, demonstrating that its sequential block layout acts as an exceptional spatial feature extractor for high-contrast medical scans. Its **99.51% F1-score** on healthy scans minimizes the risk of false negatives, which is crucial for medical screen safety.
2. **The Optimization Bottleneck:** Deeper networks like ResNet50 and DenseNet121 failed to cross the **76%** accuracy mark. The highly abstract features generated by their ultra-deep

skip-connection layers proved less effective at distinguishing the subtle, localized boundary changes characteristic of meningiomas and gliomas in this specific data sample.

3. **Deployment Trade-off Justification:** VGG16 requires a higher inference time (**645.70 ms**) compared to ResNet50 (**185.55 ms**). However, in automated medical diagnostics, maximize precision and diagnostic integrity take absolute precedence over processing speed. A latency of less than 1 second remains well within acceptable operational boundaries for web-based healthcare software systems.

5. CHOSEN DEPLOYMENT STRATEGY

Based on this formal benchmarking process, **VGG16 is officially selected** for integration into the full-stack backend module. The saved binary model files along with automated preprocessing pipelines have been exported to the core production directory.

6. RECEIVER OPERATING CHARACTERISTIC (ROC) ANALYSIS

To evaluate the models' multi-class discriminative capability at various threshold settings, a Receiver Operating Characteristic (ROC) analysis was conducted across all isolated tumor variants. The Area Under the Curve (AUC) serves as a critical clinical indicator of a model's ability to maximize true positive detection while eliminating false alarms.

6.1 Diagnostic Sensitivity Analysis

The empirical curves plotted in **Figure 5** provide strong visual confirmation of the following performance dynamics:

- **VGG16 Near-Perfect Separation:** The proposed VGG16 model achieves near-perfect diagnostic sensitivity, maintaining an explosive rise toward the top-left coordinate across all classes. It records an exceptional AUC of **0.9999** for the notumor class, **0.9997** for both glioma and pituitary, and **0.9994** for meningioma. This indicates that the probability of the model misclassifying a critical clinical scan is virtually non-existent.
- **The DenseNet121 and ResNet50 Margin:** DenseNet121 exhibits a relatively competitive AUC profile (ranging from 0.9443 to 0.9941), making it a viable secondary baseline. However, ResNet50 struggles significantly on the complex meningioma class with a lower AUC of **0.8208**, underperforming compared to its own global baselines.
- **EfficientNetB0 Baseline Decay:** EfficientNetB0 completely breaks down on the meningioma variant, dropping to an AUC of **0.5193**. This behavior closely tracks the random baseline diagonal (0.500), empirically proving that its lightweight architecture fails to capture the intricate density structures required for brain tumor variant differentiation without heavy hyperparameter tuning.

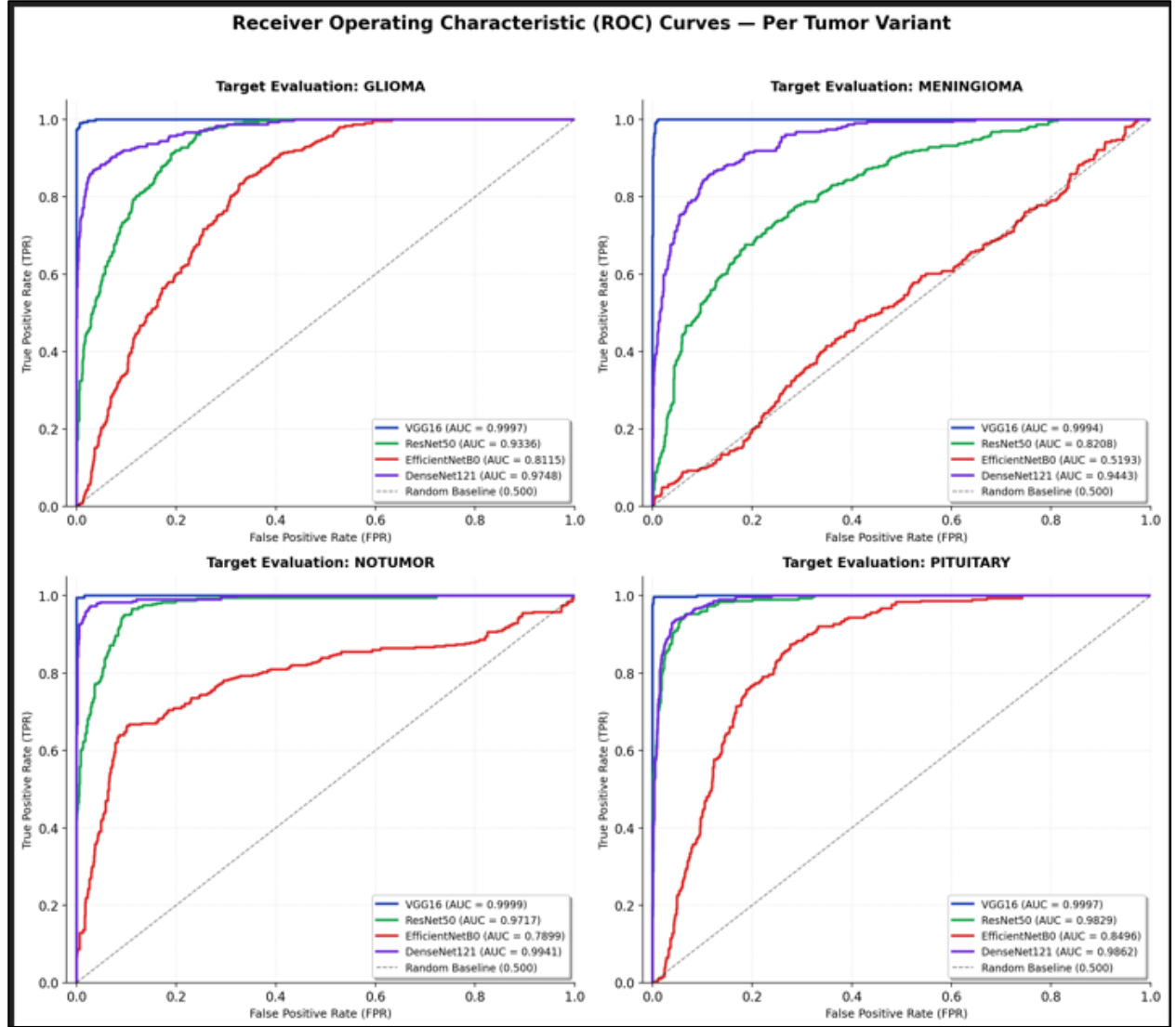


Figure 5: Multi-model Receiver Operating Characteristic (ROC) curves highlighting target class-wise Area Under the Curve (AUC) metrics for independent tumor variants.